Enterprise Meeting Intelligence Platform

System Evolution, Upgraded Feature Specifications & Architecture Report

Architecture: FastAPI + Next.js 14 + Whisper ASR	**Version:** 3.0-Production-Enterprise	**Security:** SHA-256 Chained Audit Ledger
Diarization Engine: 8-Signal / 124-Dimensional	**Test Status:** 23/23 Pytest Suites Passing	**Biometric Policy:** Sanitized / GDPR Compliant

1. Chronological Development & Evolution Timeline

• Phase 1: Foundation & ASR Latency Acceleration

Fixed Apple Silicon PyTorch MPS tensor device mismatch bug (slow_conv2d_forward_mps). Reconfigured Whisper ASR execution pipeline to optimized CPU (openai/whisper-tiny), slashing processing latency from 4 minutes down to 1–2 seconds per chunk.

• Phase 2: Real-Time Streaming & Live Personalized Minutes

Added real-time 3s WebM chunk ingestion (/audio/recording/{id}/stream) with instant LiveTranscript.tsx rendering. Overhauled NLP parser to extract sentence-level exact quotes, attendee names, financial amounts ($50k, 5 Lakhs), and added verify_faithfulness grounding accuracy score.

• Phase 3: Multi-Signal Diarization Engine (v2.5)

Engineered 124-dimensional acoustic vectors (20 MFCCs + Delta + Delta2 + Spectral Centroid + ZCR) across 0.8s sub-second sliding windows with Cosine Agglomerative Clustering and 1-click global speaker renaming.

• Phase 4: Speaker Voice Enrollment & Calibration Engine (v3.0)

Implemented 3-sample 5–10s voice calibration with dynamic anti-spoof phrases, SNR/clipping/VAD quality scorecard, biometric consent controls with 1-click vector purging, and 8th signal integration (Calibrated Identity Match).

• Phase 5: Enterprise Governance & Profile Health Dashboard

Created /voice-profiles health dashboard, 'Why This Speaker?' multi-signal explanation modal, profile drift guardrails (>0.35 distance), human review learning candidates, and 22 controlled benchmark scenarios.

2. Detailed Upgrade Breakdown: Before vs. After

Platform Feature	Initial State	Upgraded Production Capability (v3.0)
ASR Audio Transcription	3–4 min latency; PyTorch MPS tensor mismatch hangs	1–2 sec real-time local CPU execution; Bhashini Indian language integration
Live Streaming	Batch upload after meeting finished	Real-time 3s WebM chunk streaming with live synchronized transcript feed
Meeting Minutes Generation	Generic repetitive templates	Personalized sentence-level extraction of quotes, exact $ / Lakhs, and decisions
AI Faithfulness Guardrail	None (potential hallucination)	Token overlap verification with grounding_accuracy_score flagging

Speaker Diarization	Basic clustering; false turn switching	8-signal matrix with 124-dim spectral vectors over 0.8s sub-second windows
Speaker Voice Enrollment	Not available	3-sample multi-criteria voice calibration with dynamic anti-spoof phrases
Identity Classification	Forced closest assignment	4-tier matrix: MATCHED, AMBIGUOUS, UNKNOWN (never forces false match), UNAVAILABLE
Biometric Privacy & Consent	Uncontrolled	Explicit consent, sanitized public DTOs, and 1-click vector purging
Governance & Audit	Basic event logs	Cryptographic SHA-256 chained audit ledger with tamper detection

3. Production Architecture & Multi-Signal Identity Engine

The system separates **Speaker Change Confidence** from **Speaker Identity Confidence**, combining 8 independent acoustic and biometric signals into an enterprise decision matrix:

- **Signal 1–3: Spectral Acoustic Embeddings** — 20 MFCCs, Delta (velocity), and Delta-Delta (acceleration) tracking vocal tract resonance.
- **Signal 4: Pitch Statistics (F0)** — Tracks fundamental frequency mean, standard deviation, and dynamic pitch range across turns.
- **Signal 5: Spectral Centroid & ZCR** — Measures voice brightness and zero-crossing rate to separate voiced speech from unvoiced fricatives.
- **Signal 6: RMS Energy & Dynamics** — Tracks volume transitions and pauses across segment boundaries.
- **Signal 7: Voice Activity Detection (VAD)** — Filters background room noise and aligns turn boundaries with natural pauses.
- **Signal 8: Calibrated Identity Match Evidence** — Evaluates Cosine similarity against enrolled participant voice profiles.

4. Quality Assurance & Production Verification Metrics

Verification Metric	Result / Coverage	Compliance Status
Pytest Backend Test Suite	23 / 23 unit & integration tests passing	**100% Passed**
Controlled Benchmark Suite	22 scenarios (case_01 to case_22)	**Verified**
Next.js Production Build	14 / 14 static & dynamic routes compiled	**0 Errors**
ASR Processing Latency	1–2 seconds per 3-second audio chunk	**Real-Time**
Biometric Data Isolation	Raw vectors sanitized from all public APIs	**GDPR / ISO 27001**